Computer Software: Handouts

Topics: System Software, Application Software, Installing & Running

1. Introduction to Computer Software

A computer system fundamentally consists of two primary components: Hardware and Software. Hardware refers to the physical, tangible parts of the computer, such as the monitor, keyboard, and processor. However, hardware cannot function independently.

Software is a systematic collection of digital instructions, data, and programs used to operate computers and execute specific tasks. It serves as the guiding logic that directs the physical components on how to process information. Without software, the hardware remains inactive and non-functional.

Example: Hardware is analogous to a television set, while software represents the broadcasted channels and media programs displayed on it. Without the media (software), the television (hardware) serves no practical purpose.

2. System Software

System software is a classification of computer programs designed to operate the computer's hardware and provide a platform for running application software. It acts as an essential intermediary (interface) between the user, the application programs, and the physical hardware devices.

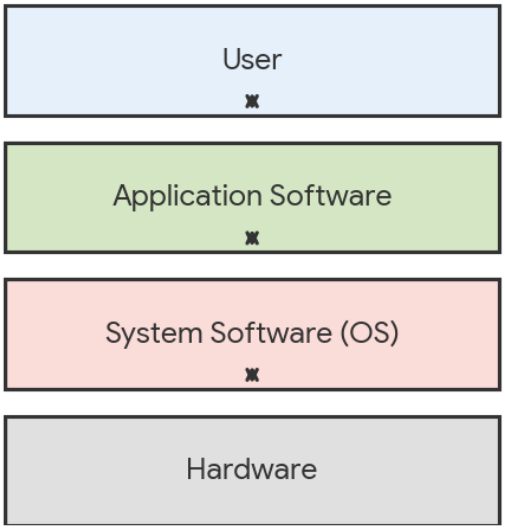
Its primary responsibility is to manage system resources, such as memory allocation, processing time, and peripheral devices, ensuring that all components work together efficiently.

Example: In a large restaurant, the system software functions similarly to the restaurant manager. The manager oversees the facility, ensures utilities are functioning, and coordinates the staff, creating the necessary environment for the chefs and servers to perform their specific duties.

Primary Categories of System Software:

- **Operating System (OS):** The core software that manages all hardware and software resources (e.g., Microsoft Windows, macOS, Linux, Android).
- **Device Drivers:** Specialized programs that enable the Operating System to communicate with specific hardware devices (e.g., printer drivers, display drivers).
- **Utility Software:** Diagnostic and maintenance tools that optimize system performance (e.g., Antivirus software, Disk formatting tools).

Figure 1: Interaction between User, Software, and Hardware



3. Application Software

Application software, commonly referred to as an 'App,' is designed to perform specific, user-oriented tasks. Unlike system software, which operates primarily in the background to manage resources, application software is utilized directly by the end-user to accomplish practical objectives, such as document creation, data analysis, or entertainment.

Example: Continuing the restaurant analogy, if the system software is the manager, the application software is the chef. The chef is hired to execute a highly specific task—preparing meals for the customers.

Primary Categories of Application Software:

- **Word Processors:** Software used for creating, editing, and formatting text documents (e.g., Microsoft Word, Google Docs).
- **Web Browsers:** Applications designed for accessing and navigating the World Wide Web (e.g., Google Chrome, Mozilla Firefox).
- **Multimedia Software:** Tools for playing, editing, or creating audio and video content (e.g., VLC Media Player, Adobe Premiere).

4. Comparison: System vs. Application Software

Characteristic	System Software	Application Software
Primary Function	Manages system resources and hardware.	Performs specific tasks for the end-user.
Dependency	Operates independently and is required to boot the system.	Requires System Software (OS) to function.

Characteristic	System Software	Application Software
User Interaction	Minimal direct user interaction; runs in the background.	High degree of direct user interaction.
Examples	Windows 11, Linux, Device Drivers.	Microsoft Word, Web Browsers, Media Players.

5. Installing and Running Software

Software Installation Process

Software distribution packages are typically compressed into single executable files (e.g., .exe, .msi, .apk). Installation is the required procedure of extracting these compressed files, organizing them into the appropriate directories on the storage drive, and registering the software's configurations with the Operating System.

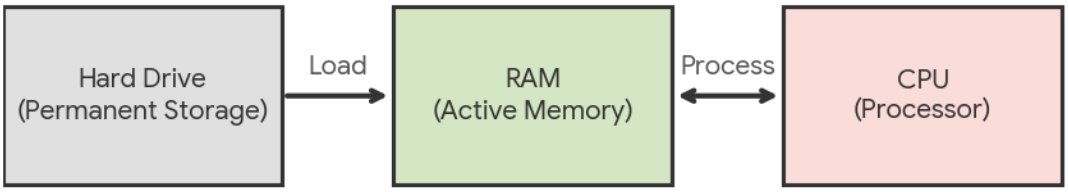
Example: Purchasing a disassembled piece of furniture. It arrives in a compact box, but it cannot be utilized immediately. The assembly process—unpacking, putting the pieces together, and placing it in a room—is analogous to the software installation process.

Executing (Running) Software

Once installed, software resides permanently on the system's Hard Drive. However, the Central Processing Unit (CPU) cannot efficiently execute instructions directly from standard storage drives due to speed limitations. When a user initiates (runs) an application, the system copies the necessary data from the Hard Drive into the Random Access Memory (RAM). RAM is volatile but provides the high-speed access required by the CPU.

Example: Consider a hard drive as a large library bookshelf where information is permanently stored, and RAM as a study desk. To read a book efficiently, one must retrieve it from the shelf (Hard Drive) and place it on the desk (RAM). When finished, the book is closed and returned to the shelf.

Figure 2: Software Execution Flow



6. Essential Terminology

- **Hardware:** The physical and tangible components of a computing system.
- **Software:** The intangible programs and data that govern the operation of a computer.
- **RAM (Random Access Memory):** Volatile primary memory used to temporarily store data that the CPU is actively processing.
- **Storage (Hard Drive/SSD):** Non-volatile secondary memory utilized for the permanent retention of data and installed applications.
- **Interface:** A point of interaction between discrete systems. A User Interface (UI) enables human-computer interaction, while software interfaces allow applications to communicate with the OS.
- **CPU (Central Processing Unit):** The primary processor that executes logical and arithmetic instructions dictated by software.